INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Areas, Shapes, Perimeter & Mensuration

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Areas, Shapes, Perimeter & Mensuration

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/areas-shapes-perimeter/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4

📌 Section 1: Rules & Core Concepts to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Triangle: Area = 1/2 * base * height; Heron's formula = sqrt(s(s-a)(s-b)(s-c)) where s = (a+b+c)/2.
- Equilateral Triangle: Area = (sqrt(3)/4) * a^2; Height = (sqrt(3)/2) * a.
- Circle: Area = pi * r^2; Circumference = 2 * pi * r.
- Cylinder: Volume = pi * r^2 * h; Curved Surface Area = 2 * pi * r * h; Total Surface Area = 2 * pi * r * (r + h).
- Cone: Volume = (1/3) * pi * r^2 * h; Slant height l = sqrt(r^2 + h^2); Curved Surface Area = pi * r * l.
- Sphere: Volume = (4/3) * pi * r^3; Surface Area = 4 * pi * r^2.
- HCF and LCM

```
Questions

Formulas

How to Solve Quickly

Shortcuts, Tips & Tricks
```

  - Number System

```
Questions

Formulas

How to Solve Quickly

Shortcuts, Tips & Tricks
```

- Number Decimals & Fractions

```
Questions

Formulas

How to Solve Quickly

Shortcuts, Tips & Tricks
```

  - Surds and Indices

```
Questions

Formulas

How to Solve Quickly

Shortcuts, Tips & Tricks
```

    - Divisibility

```
Questions

Formulas

How to Solve Quickly

Shortcuts, Tips & Tricks
```

      - Ages

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
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- LCM

```
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```

  - HCF

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

    - Inverse

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

      - Speed Time and Distance

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

        - Work and Time

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

- Boats and Streams

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

⚡ Section 2: How to Solve Quickly (Exam Elimination Strategies)

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🎯 Section 3: High-Yield Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Percentage change in area when dimensions change: If radius increases by r%, area increases by [2r + (r^2/100)]%.
- If ratio of radii of two circles is a:b, ratio of their areas is a^2 : b^2.
- HCF and LCM

```
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📝 Section 4: Comprehensive PrepInsta Practice Question Bank (All Quizzes)

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
The length of a rectangular plot is 20 meters more than its breadth. If the cost of fencing the plot @ 26.50 per meter is Rs. 5300, what is the length of the plot in meters?

Options:
- **(A)** `40`
- **(B)** `50`
- **(C)** `120`
- **(D)** `Data inadequate`
- **(E)** `None of these`

Correct Answer: (E) `None of these`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let breadth = x metres.



Then, length = (x + 20) metres.



Perimeter = ( 5300/26.50 ) m = 200 m.



2[(x + 20) + x] = 200



2x + 20 = 100



2x = 80



x = 40.



Hence, length = x + 20 = 60 m.
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
A rectangular field is to be fenced on three sides leaving a side of 20 feet uncovered. If the area of the field is 680 sq. feet, how many feet of fencing will be required?

Options:
- **(A)** `34`
- **(B)** `40`

- **(C)** `68`
- **(D)** `88`

Correct Answer: (D) `88`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
We have: l = 20 ft and lb = 680 sq. ft.




So, b = 34 ft.




Length of fencing = (l + 2b) = (20 + 68) ft = 88 ft.
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:
A tank is 25 m long, 12 m wide and 6 m deep. The cost of plastering its walls and bottom at 75 paise per sq. m, is:

Options:
- **(A)** `Rs. 456`
- **(B)** `Rs. 458`
- **(C)** `Rs. 558`
- **(D)** `Rs. 568`

Correct Answer: (C) `Rs. 558`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Area to be plastered =



[2(l+b)×h]+(l×b)



=[2(25+12)×6]+(25×12)



=744sq.m



Cost of plastering = 744×75100 = Rs.558
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:
A rectangular park 60 m long and 40 m wide has two concrete crossroads running in the middle of the park and rest of the park has been used as a lawn. If the area of the lawn is 2109 sq. m, then what is the width of the road?

Options:
- **(A)** `2.91 m`
- **(B)** `3 m`
- **(C)** `5.82 m`
- **(D)** `None of these`

Correct Answer: (B) `3 m`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Area of the park = (60 x 40) m2 = 2400 m2.



Area of the lawn = 2109 m2.



Area of the crossroads = (2400 - 2109) m2 = 291 m2.



Let the width of the road be x metres. Then,



60x + 40x - x2 = 291



=&gt;x2 - 100x + 291 = 0



=&gt;(x - 97)(x - 3) = 0



As, the width of the road cannot exceed the length of rectangular park,



So, x = 3.
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:

A rabbit is tied to one end of an equilateral triangle of side 5 m with a rope length of 8 m.The rabbit is not

allowed to travel inside the triangle then find the maximum area covered by the rabbit?

Options:

- **(A)** `180.24 m2`
- **(B)** `190.24 m2`
- **(C)** `190.44 m2`
- **(D)** `190 m2`

Correct Answer: (B) `190.24 m2`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Area covered by rabbit = Area of circle(radius 8m)-


Area of Triangle(side 5m)


= π*8*8 - (√(3)/4)*5*5


= 201.06 - 10.82


= 190.24 m2
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:

No. of triangles formed in a polygon having 15 sides is?

Options:

- **(A)** `400`
- **(B)** `355`
- **(C)** `353`
- **(D)** `455`

Correct Answer: (D) `455`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
total no. of points=15


no. of points needed to form a triangle=3


thus total no of triangles formed=^{15}C_{3}=455.
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:
A GOLF BALL HAS DIAMETER EQUAL TO 4.1CM. ITS SURFACE HAS 150 DIMPLES EACH OF RADIUS 2MM (0.2 CM). CALCULATE TOTAL SURFACE AREA WHICH IS EXPOSED TO SURROUNDINGS ASSUMING THAT THE DIMPLES ARE HEMISPHERICAL.

Options:
- **(A)** `51.62 cm2`
- **(B)** `77.62 cm2`
- **(C)** `71.62 cm2`
- **(D)** `45.62 cm2`

Correct Answer: (C) `71.62 cm2`

PrepInsta Detailed Explanation & Mathematical Derivation:

Given the diameter of golf ball = 4.1 cm

=> the radius of golf ball

= diameter/2 = 4.1/2cm

Now surface area of a golf ball without dimples

4 *π r2 = 4 * π * (4.1/2)2 = 16.81 π cm2

It is given that the shape of dimples is hemispherical and radius of each dimple = 2mm = 0.2 cm

Area of each dimple =

= 2 πr2 = 2 * π * (0.2)2 = 0.08 π cm2

So area of 150 dimples = 150 x area of each dimple = 150 x 0.8π = 12π cm square

Area of flat surface removed to make 1 dimple

```
=  πr2 = π * (0.2 )2 = 0.04 π cm2



Area of flat surface removed to make 150 dimples



= 150 * 0.04 π = 6 π cm2



Now total surface area of golf ball exposed to surroundings = surface area of golf ball without dimples + area of 150 dimples – area of flat surface removed to make 150 dimples



= 16.81 π + 12π - 6π



= 22.81π



= 22.81 x 3.14



= 71.62 cm2
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:

If a goat is tied to a pole at Point A with a rope 12 m long such that it can not enter a triangle ABC with AB=AC=10 m and angle A = 30 deg. How much area can it graze?

Options:

- **(A)** `(i) less than 132 π`
- **(B)** `(ii) more than 132 π`
- **(C)** `(iii) equal to 132 π`
- **(D)** `(iv) none`

Correct Answer: (B) `(ii) more than 132 π`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Since the length of the rope is longer than the sides of the triangle it


will eat a bit more at the edges than (330/360)*144π. Hence more than


132π.
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:

The right-angled triangle PQR is to be constructed in the xy-plane so that the right angle is at P and PR is parallel to the x-axis. The x and y coordinates of P, Q, and R are to be integers that satisfy the inequality $-4 \le x \le 5$ & $6 \le y \le 16$. How many different triangles with these properties could be constructed?

Options:

- **(A)** `A. 1,100`
- **(B)** `B. 12,100`
- **(C)** `C. 10,000`
- **(D)** `D. 9,900`

Correct Answer: (D) `D. 9,900`

PrepInsta Detailed Explanation & Mathematical Derivation:

We have the rectangle with dimensions 10*11

(10 horizontal dots and 11 vertical). PQ is parallel to y-axis and PR

is parallel to x-axis.

Choose the (x,y) coordinates for vertex P (right angle):

$^{10}C_{1}\times ^{11}C_{1}$;

Choose the x coordinate for vertex R (as y coordinate is fixed

by

A): $^{9}C_{1}$, (10-1=9 as 1 horizontal dot is already occupied by A);

Choose the Y coordinate for vertex Q (as x coordinate is

fixed by A):

10C1, (11-1=10 as 1 vertical dot is already

occupied by A).

```
^{10}C_{1}\times ^{11}C_{1}\times ^{9}C_{1}\times ^{10}C_{1}=9900.
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:
If PQRST is a parallelogram what it the ratio of triangle PQS & parallelogram PQRST

Options:
- **(A)** `1:2`
- **(B)** `2:1`
- **(C)** `1:3`
- **(D)** `3:1`

Correct Answer: (A) `1:2`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
1:2


as traingle resides half of parallelogram
```
